The JPEG Still Picture Compression Standard

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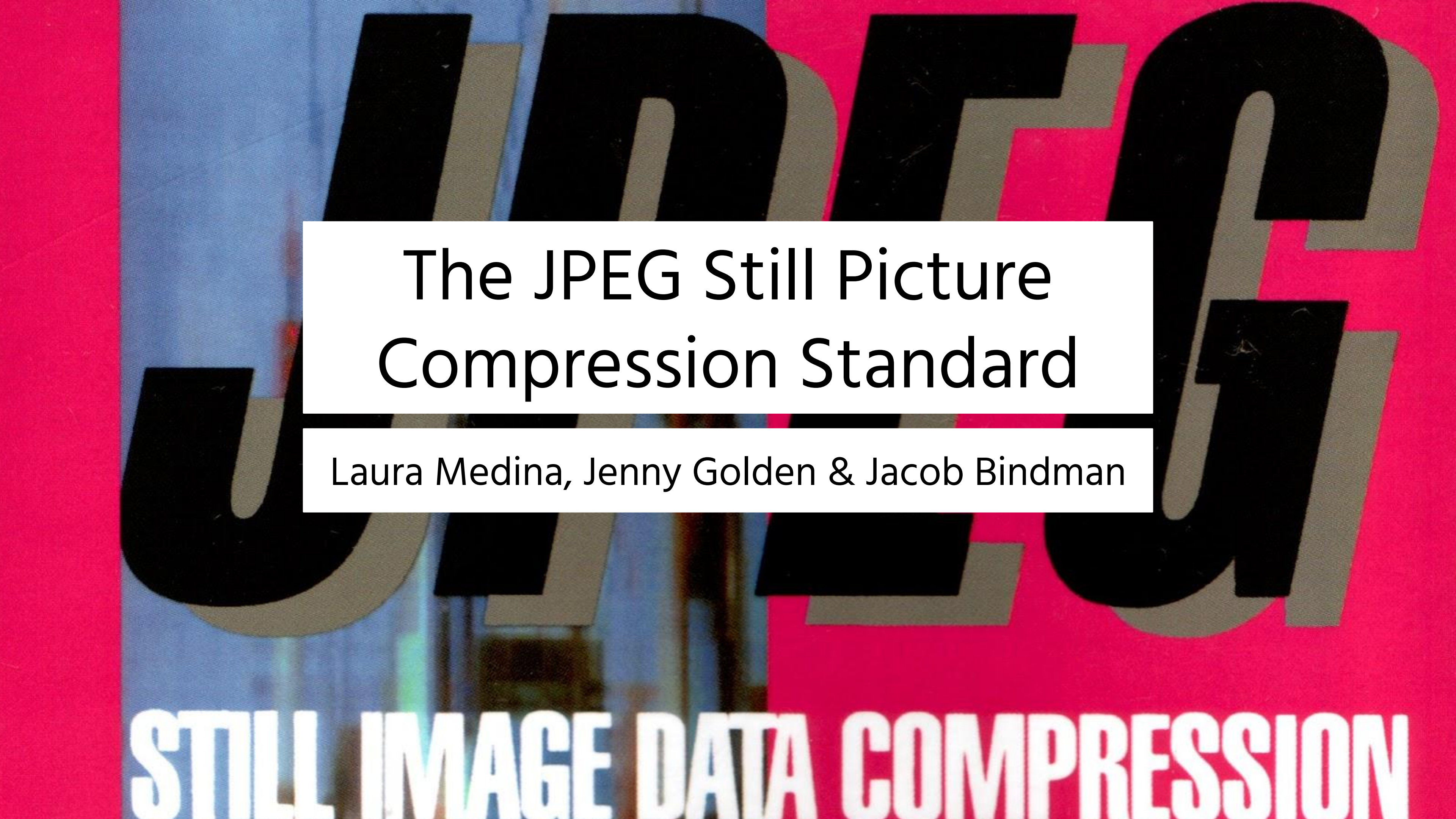


ADVANCES

1992

- Great advances in digital technology
- Limitations of data transfer

GOALS

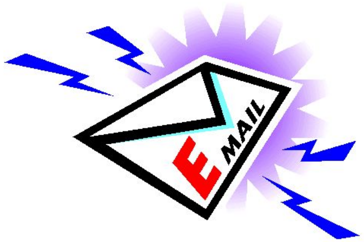
JPEG

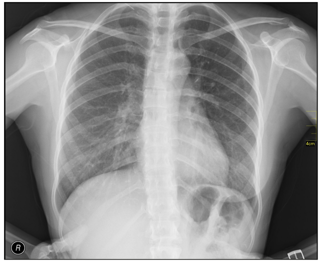
Joint

Photographic

Experts

Groups





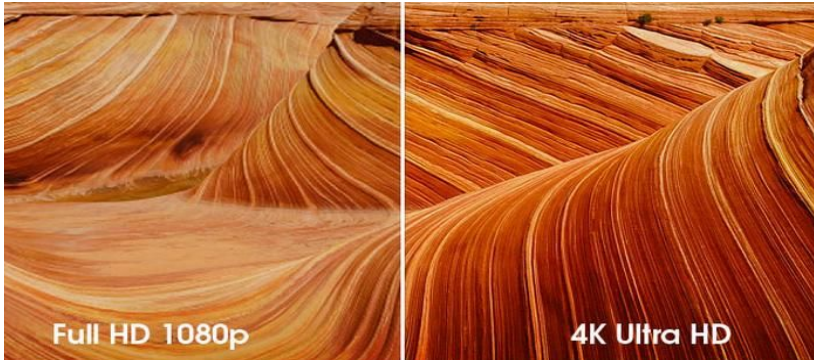
METHODS

- 12 Methods proposed for image-compression standard
  - Based on picture quality
  - Narrowed down to 3

- After analyzing the three, DCT had produced best picture quality

GOALS

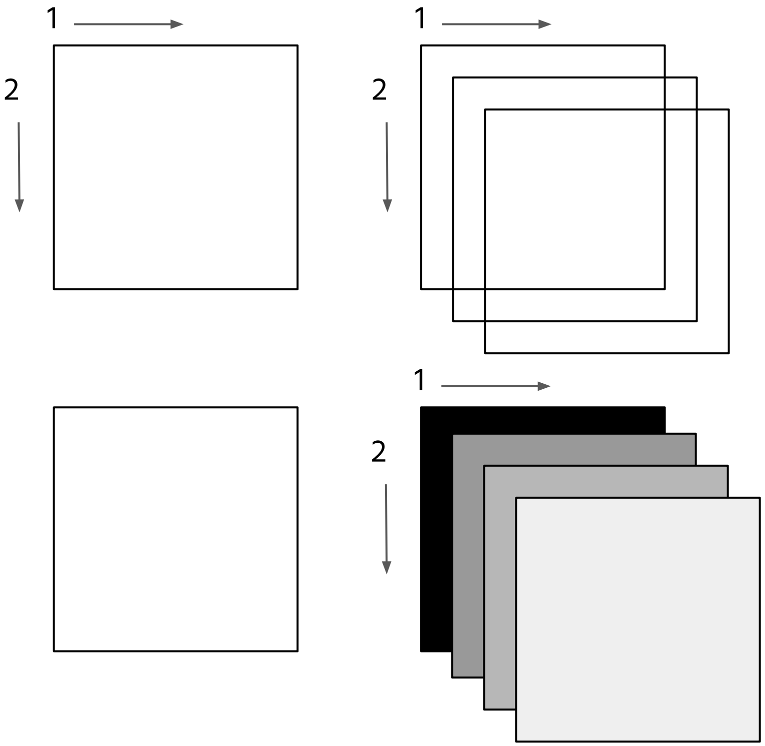
- Allow user to create tradeoff between quality and size; not 1:1 algorithm
- Targets full spectrum of all image compression (medical, personal, etc)
- Widespread use: can compress many images regardless of color, pixel aspect ratio, complexity, etc
- Very accessible



GOALS

4 Modes of operation:

- Sequential (left to right, top to bottom)
- Progressive (multiple passes)
- Lossless (encoded to exact recovery, no minimization of file size)
- Hierarchical (encoded in multiple resolutions)



PAPER

- 2 basic methods
- DCT(Discrete Cosine Transformation): Lossy
- Predictive: Lossless

- This paper focuses mostly on Baseline DCT which is a subset of DCT.
    - Baseline is simple lossy technique, widely implemented by jpeg

DCT-BASED

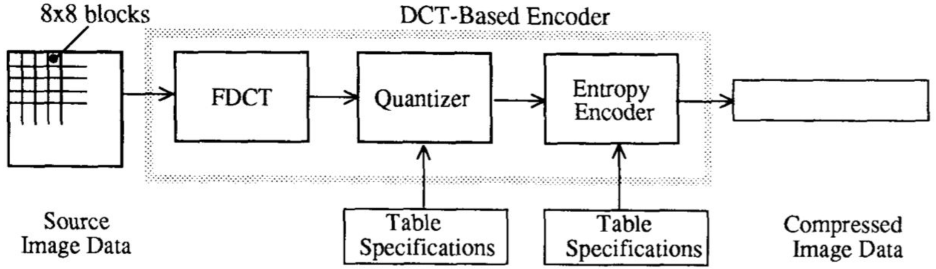
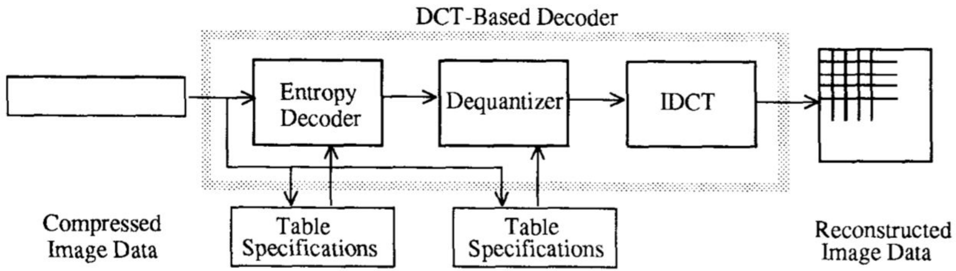


Figure 1. DCT-Based Encoder Processing Steps



BACKGROUND PROBLEM DESIGN **EXPERIMENT** SOLUTION FUTURE

FDCT ENCODER

$$F(u,v) = \frac{1}{4} C(u)C(v)\left[\sum_{x=0}^{7}\sum_{y=0}^{7} f(x,y) * \cos\frac{(2x+1)u\pi}{16}\cos\frac{(2x+1)v\pi}{16}\right] \quad (1)$$

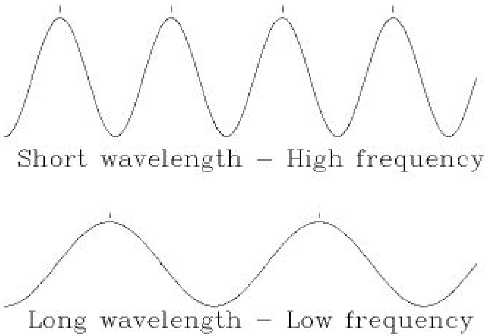
encoder

$$f(x,y) = \frac{1}{4}\left[\sum_{u=0}^{7}\sum_{v=0}^{7} C(u)C(v)F(u,v) \cos\frac{(2x+1)u\pi}{16}\cos\frac{(2x+1)v\pi}{16}\right] \quad (2)$$

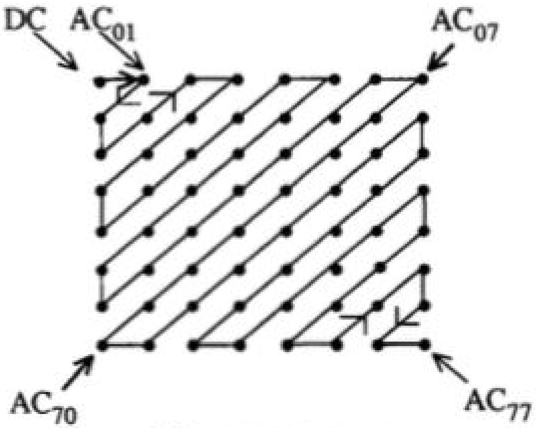
decoder

where: $C(u),\ C(v) = 1/\sqrt{2}$ for $u,v = 0$;

$C(u),\ C(v) = 1$ otherwise.



QUANTIZER



Zig-zag sequence

HIERARCHY

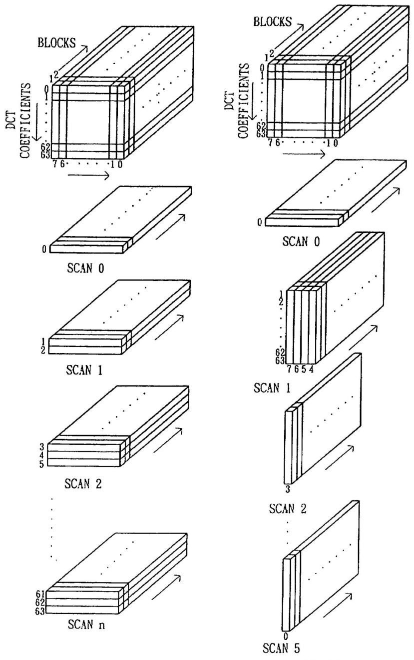


FIG. 1 (PRIOR ART)

FIG. 2 (PRIOR ART)

- Encoding of single image at multiple resolutions
- Adjacent encodings differ by multiple of 2
- 5 steps of encoding procedure
  - Filter and downsample in multiples of 2
  - Encode this reduced-size image
  - Decode reduced-size image and then interpolate and un-sample by 2 using identical interpolation filter
  - Use the obtained unsampled image as prediction of original, and encode the image difference
  - Repeat 3 and 4 until fully encoded

IMPROVEMENTS

JPEG

JPEG 2000

JPEG

JPEG 2000

- More efficient
- Possibility of lossless compression
- Noise resilience
- Better coding and decoding process through many different cycles

RESOURCES

CS179

Design of Useful and Usable Interactive Systems

CS171

Visualization

CS175

Computer Graphics

WiCS
WOMEN IN COMPUTER SCIENCE

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